Assignment 2: FPGA Implementation

Fall 2026

1 Scenario

During this assignment, you will again play the role of an engineer working at a market maker. However, this time, the hot path is implemented on an FPGA. The pipeline is described in `design.pdf`, but is not yet complete.

The underlying trading strategy is the same as in Assignment 1, with minor variations to account for the difference in platforms.

2 Deliverables

For this assignment, you will be asked to turn in the completed codebase and a 1-page write-up.

- Completed codebase
  - Complete the `market_data_parser`
  - Complete the `order_encoder`
  - Complete the `sequence_check`
  - Complete the `book`
  - Modify one of the other modules so that we can increase the clock frequency. Ideally, you should be able to raise the clock frequency from under 20 MHz to over 30 MHz. `make timing` is not perfectly deterministic: a difference of a few percent between two runs of the same design is placement noise, not a real change, so judge your improvement by the overall trend rather than a single run.
- Write-up
  - Explain the change you made and how it allows us to increase the clock frequency.
  - Explain how your implementation of the order book differs from the C++ implementation of the order book. How does the implementation of each order book play to the strengths of the respective platform?

Please use the provided `write-up.tex` format and submit the resulting PDF. You are permitted (and encouraged) to collaborate with other students and use AI tools to help you understand the application, evaluate proposed changes, etc. That said, you should write the final 1-page write-up yourself, and the interview will be conducted without any outside help, so take time to make sure that you also understand the application and your write-up answers.

Begin your submission with a short statement describing how you collaborated with other students, which AI tools you used, and how you used them. This statement does not count against the 1-page limit for the write-up.